MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE S2S

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated AWG no. 18, class D stranding. Maximum diameter 0.050 inch.
Second - Cross-linked thermosetting polyethylene insulation, colored black. Maximum wall thickness 0.060 inch.
Third - Braided shield of AWG no. 34 tin coated copper strands. Sixteen carriers, six ends per carrier, eight picks per inch.
Fourth - Synthetic rubber insulation. Nominal wall thickness 0.055 inch.
Fifth - Braided shield consisting of 0.0063-inch diameter tin-coated, copper-clad steel wires, class 40A in accordance with ASTM B 452. Twenty-four carriers, eight ends per carrier, nine or ten picks per inch.
Sixth - Two-layer jacket with the layers bonded together, with a reinforcing braid between layers, and having a total thickness of 0.065-inch minimum. Inner layer of synthetic rubber insulation having 0.033-inch nominal wall thickness. Outer layer of arctic type polychloroprene. Reinforcing braid of 210-denier nylon, applied with 12 carriers, 2 ends per carrier and 3 picks per inch. Completed cable overall diameter 0.480-inch minimum, 0.500-inch maximum.
Seventh - Cable surface marking required.

FSC 6145

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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - ohms/1000 feet, at 25°C, maximum. | 7.05 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to inner shield..... | 1500 |
| Shield to shield..... | 1500 |
| Insulation resistance - megohms/1000 feet, minimum | |
| Conductor to inner shield..... | 500 |
| Shield to shield..... | 100 |
| Outer shield to water..... | 10 |
| Group A: | |
| Visual and dimensional..... | No failure |
| Hydrostatic (open end) (test every shipping length) - at a gauge pressure of 1000 lbf/in$^2$, for 6 hours, leakage. | Zero |
| Capacitance - at 1 kHz, pF/ft, maximum..... | 37 ± 1 |
| Group B: | |
| Breaking strength - pounds, minimum..... | 150 |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - lbf/in$^2$, minimum..... | 1500 |
| Elongation - percent, minimum..... | 250 |
| Jacket (outer layer) | |
| Tensile strength - lbf/in$^2$, minimum..... | 1800 |
| Elongation - percent, minimum..... | 300 |
| Set - inch, maximum..... | 3/8 |
| Drip - at 95 ± 1°C..... | Zero |
| Group C: | |
| Flammability - inches, maximum..... | 2 |
| Physicals (aged) | |
| Air pressure heat | |
| Insulation | |
| Tensile strength - percent of unaged, minimum..... | 50 |
| Elongation - percent of unaged, minimum..... | 50 |
| Permanence of printing (jacket) - cycles, minimum..... | 250 |
| Cable filler removability..... | No failure |
| Heat distortion - percent of unaged, maximum..... | 30 |
| Shield-conformance to material coverage and construction..... | No failure |



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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following: Pressure cycling test in accordance with 4.7.25, except as follows:

(1) The specimen shall consist of a 10-foot length of completed cable.
(2) Instrumentation shall be connected to one of the protuding cable ends so that capacitance can be measured continuously during the test.
(3) Hydrostatic pressure in the chamber shall be raised from atmospheric pressure to a gauge pressure of 1000 $lbf/in^2$ and held for 1 hour. The specimen shall then be given 10 cycles from a gauge pressure of 1000 $lb/in^2$ to atmospheric to a gauge pressure of 1000 $lbf/in^2$, with a 10-second dwell time at each extreme. At the end of the tenth cycle, gauge pressure shall be held at 1000 $lbf/in^2$ for 1 hour.
(4) Any capacitance measurement during the test which is less than 36 pF/ft or greater than 40 pF/ft shall constitute failure.

Cable aging and compatibility (125 $\pm$ 5°C)..... No failure

UNIT ORDERING LENGTH: 500 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0885-56)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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